使用ONTAP REST API管理功能

ONTAP Automation

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使用ONTAP REST API管理功能

利用REST API管理EMS事件通知ONTAP

您可以使用ONTAP Sesrest API來設定事件管理系統（EMS）事件通知傳送給您的重要性、以及擷取具有特定屬性的所有EMS訊息或特定訊息。您可以使用REST API來協助自動化與EMS相關的工作、節省使用其他EMS訊息方法的時間。

以下範例說明如何使用ONTAP Resrest API來處理EMS訊息。在每個範例中、請務必在方括弧<>中指定資訊的值。您也可以參考範例 "Python指令碼" 這說明如何將許多與EMS相關的活動自動化。

檢視特定事件記錄

使用「Get /support/ES/events」API呼叫、您可以擷取特定事件訊息、例如最新訊息、包含特定文字的訊息、或是具有特定嚴重性的訊息。

申請範例：擷取最新訊息

```
curl -X GET -u admin:<PASSWORD> -k
'https://<IP_ADDRESS>/api/support/ems/events?fields=message.name&max_recor
ds=1'
```

回應範例


```
"records": [
    {
      "node": {
        "name": "malha-vsim1",
        "uuid": "da4f9e62-9de3-11ec-976a-005056b369de",
        "_links": {
          "self": {
            "href": "/api/cluster/nodes/da4f9e62-9de3-11ec-976a-
005056b369de"
          }
        }
      },
      "index": 6260,
      "message": {
        "name": "netinet6.rtr.high.mtu"
      },
      "_links": {
        "self": {
          "href": "/api/support/ems/events/malha-vsim1/6260"
        }
      }
    }
  ],
  "num_records": 1,
  "_links": {
    "self": {
      "href": "/api/support/ems/events?fields=message.name&max_records=1"
    },
    "next": {
      "href": "/api/support/ems/events?start.keytime=2022-03-
22T10%3A28%3A29-04%3A00&start.node.name=malha-
vsim1&start.index=6260&fields=message.name&max_records=1"
    }
  }
}
```


申請範例：擷取內含特定文字和嚴重性的訊息

```
curl -X GET -u admin:<PASSWORD> -k
'https://<IP_ADDRESS>/api/support/ems/events?log_message=*disk*&message.se
verity=alert'
```

回應範例


```
"records": [
    {
      "node": {
        "name": "malha-vsim1",
        "uuid": "da4f9e62-9de3-11ec-976a-005056b369de",
        "_links": {
          "self": {
            "href": "/api/cluster/nodes/da4f9e62-9de3-11ec-976a-
005056b369de"
          }
        }
      },
      "index": 4602,
      "time": "2022-03-18T06:37:46-04:00",
      "message": {
        "severity": "alert",
        "name": "raid.autoPart.disabled"
      },
      "log_message": "raid.autoPart.disabled: Disk auto-partitioning is
disabled on this system: the system needs a minimum of 4 usable internal
hard disks.",
      "_links": {
        "self": {
          "href": "/api/support/ems/events/malha-vsim1/4602"
        }
      }
    }
  ],
  "num_records": 1,
  "_links": {
    "self": {
      "href":
"/api/support/ems/events?log_message=*disk*&message.severity=alert&max_rec
ords=1"
    },
    "next": {
      "href": "/api/support/ems/events?start.keytime=2022-03-
18T06%3A37%3A46-04%3A00&start.node.name=malha-
vsim1&start.index=4602&log_message=*disk*&message.severity=alert"
    }
  }
}
```

檢視現有的EMS組態

使用「Get /support/ES/」API呼叫、您可以擷取現有EMS通知組態的相關資訊。

申請範例

```
curl -X GET -u admin:<PASSWORD> -k 'https://<IP_ADDRESS>/api/support/ems/'
```

回應範例

```
{
  "proxy_url": "https://proxyserver.mycompany.com",
  "proxy_user": "proxy_user",
  "mail_server": "mail@mycompany.com",
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "pubsub_enabled": "1",
  "mail_from": "administrator@mycompany.com"
}
```

建立新的EMS通知組態

您可以同時使用數個API呼叫來建立新的EMS組態、以自訂接收訊息的方式和時間。

步驟

1. 設定全系統的電子郵件設定。例如：

   ```
   curl -X PATCH -u admin:<PASSWORD> -k
   'https://<IP_ADDRESS>/api/support/ems?mail_from=administrator@mycompany.
   com&mail_server=mail@mycompany.com'
   ```

2. 定義使用特定規則來符合特定訊息的篩選條件。例如：

   ```
   curl -u admin:<PASSWORD> -X POST -d '{"name": "test-filter",
   "rules.type": ["include"], "rules.message_criteria.severities":
   ["emergency"]}' -k 'https://<IP_ADDRESS>/api/support/ems/filters/'
   ```

3. 建立訊息的目的地。例如：

```
curl -u admin:<PASSWORD> -X POST -d '{"name": "test-destination",
"type": "email", "destination": "administrator@mycompany.com",
"filters.name": ["important-events"]}' -k
'https://<IP_ADDRESS>/api/support/ems/destinations/'
```

不只是使用靜態API、更是使用非靜態CLI命令ONTAP ONTAP

利用此功能、您可以使用比支援許多工作的CLI更少的命令來自動化工作流程。ONTAP ONTAP例如、您可以使用單一POST API方法來建立篩選器、而非使用多個CLI命令。下表顯示完成一般EMS工作所需的CLI命令、以及對應的REST API呼叫：

靜態API ONTAP	CLI ONTAP
《Get /support/EMS》（取得/支援/EMS）	「事件組態顯示」
"POST /support/ES/DESTINS/DESTINATION"	1. 事件通知目的地建立 2. 建立活動通知
《Get /support/ES/events》（取得/支援/緊急醫療服務/活動	「事件記錄顯示」
"POST /support/ES/filer"	1. 「事件篩選器create -filter名稱<Filter name>」 2. 「事件篩選規則add -filter名稱<Filter name>」

相關資訊

- "靜態API EMS Python指令碼範例ONTAP"
- "REST API：自動通知高嚴重性事件ONTAP"

利用REST API管理DACL和SACL檔案權限ONTAP

使用系統存取控制清單（SACL）和自由存取控制清單（DACL）來指派檔案物件的權限。ONTAP從靜止9.9.1開始ONTAP 、ONTAP 包含端點、可指派SACL和DACL權限給檔案、並自動化檔案安全權限。（您可以深入瞭ONTAP 解在執行版本號之前、將SACL和DACL權限自動化的選項 "請按這裡"）。

從ONTAP 靜止9.9.1開始、您可以使用單一REST API呼叫來取代多個CLI命令或ONTAPI呼叫。下列範例說明如何使用ONTAP R靜止API來處理檔案權限。在每個範例中、請務必在方括弧<>中指定資訊的值。

您也可以參考範例 "Python指令碼" 這說明如何將許多SACL和DACL相關活動自動化。

檢視有效權限

使用「Get /傳輸 協定/file-Security /有效權限/」API呼叫、您可以擷取特定檔案或目錄的目前權限。

申請範例：

```
curl -X GET -u admin:<PASSWORD> -k
'https://<IP_ADDRESS>/api/protocols/file-security/effective-
permissions/cf5f271a-1beb-11ea-8fad-
005056bb645e/administrator/windows/%2F?share.name=sh1&return_records=true'
```

回應範例

```
{
  "svm": {
    "uuid": "cf5f271a-1beb-11ea-8fad-005056bb645e",
    "name": "vs1"
  },
  "user": "administrator",
  "type": "windows",
  "path": "/",
  "share": {
    "path": "/"
  },
  "file_permission": [
    "read",
    "write",
    "append",
    "read_ea",
    "write_ea",
    "execute",
    "delete_child",
    "read_attributes",
    "write_attributes",
    "delete",
    "read_control",
    "write_dac",
    "write_owner",
    "synchronize",
    "system_security"
  ],
  "share_permission": [
    "read",
    "read_ea",
    "execute",
    "read_attributes",
    "read_control",
    "synchronize"
  ]
}
```

檢視所有稽核資訊

使用「Get /傳輸 協定/file-Security /permissions/」API呼叫、您可以擷取特定檔案或目錄的所有稽核資訊。

申請範例：

```
curl -X GET -u admin:<PASSWORD> -k
'https://<IP_ADDRESS>/api/protocols/file-security/permissions/9479099d-
5b9f-11eb-9c4e-0050568e8682/%2Fparent'
```

回應範例


```
{
  "svm": {
    "uuid": "9479099d-5b9f-11eb-9c4e-0050568e8682",
    "name": "vs1"
  },
  "path": "/parent",
  "owner": "BUILTIN\\Administrators",
  "group": "BUILTIN\\Administrators",
  "control_flags": "0x8014",
  "acls": [
    {
      "user": "BUILTIN\\Administrators",
      "access": "access_allow",
      "apply_to": {
        "files": true,
        "sub_folders": true,
        "this_folder": true
      },
      "advanced_rights": {
        "append_data": true,
        "delete": true,
        "delete_child": true,
        "execute_file": true,
        "full_control": true,
        "read_attr": true,
        "read_data": true,
        "read_ea": true,
        "read_perm": true,
        "write_attr": true,
        "write_data": true,
        "write_ea": true,
        "write_owner": true,
        "synchronize": true,
        "write_perm": true
      },
      "access_control": "file_directory"
    },
    {
```

```
      "user": "BUILTIN\\Users",
      "access": "access_allow",
      "apply_to": {
        "files": true,
        "sub_folders": true,
        "this_folder": true
      },
      "advanced_rights": {
        "append_data": true,
        "delete": true,
        "delete_child": true,
        "execute_file": true,
        "full_control": true,
        "read_attr": true,
        "read_data": true,
        "read_ea": true,
        "read_perm": true,
        "write_attr": true,
        "write_data": true,
        "write_ea": true,
        "write_owner": true,
        "synchronize": true,
        "write_perm": true
      },
      "access_control": "file_directory"
    }
  ],
  "inode": 64,
  "security_style": "mixed",
  "effective_style": "ntfs",
  "dos_attributes": "10",
  "text_dos_attr": "----D---",
  "user_id": "0",
  "group_id": "0",
  "mode_bits": 777,
  "text_mode_bits": "rwxrwxrwx"
}
```


套用新權限

您可以使用「POST /傳輸 協定/file-Security /權限/」API呼叫、將新的安全性描述元套用至檔案或目錄。

申請範例

```
curl -u admin:<PASSWORD> -X POST -d '{ \"acls\": [ { \"access\":
\"access_allow\", \"advanced_rights\": { \"append_data\": true,
\"delete\": true, \"delete_child\": true, \"execute_file\": true,
\"full_control\": true, \"read_attr\": true, \"read_data\": true,
\"read_ea\": true, \"read_perm\": true, \"write_attr\": true,
\"write_data\": true, \"write_ea\": true, \"write_owner\": true,
\"write_perm\": true }, \"apply_to\": { \"files\": true, \"sub_folders\":
true, \"this_folder\": true }, \"user\": \"administrator\" } ],
\"control_flags\": \"32788\", \"group\": \"S-1-5-21-2233347455-2266964949-
1780268902-69700\", \"ignore_paths\": [ \"/parent/child2\" ], \"owner\":
\"S-1-5-21-2233347455-2266964949-1780268902-69304\", \"propagation_mode\":
\"propagate\"}' -k 'https://<IP_ADDRESS>/api/protocols/file-
security/permissions/9479099d-5b9f-11eb-9c4e-
0050568e8682/%2Fparent?return_timeout=0'
```

回應範例

```
{
  "job": {
    "uuid": "3015c294-5bbc-11eb-9c4e-0050568e8682",
    "_links": {
      "self": {
        "href": "/api/cluster/jobs/3015c294-5bbc-11eb-9c4e-0050568e8682"
      }
    }
  }
}
```

更新安全性描述元資訊

您可以使用「修補程式/傳輸協定/檔案安全性/權限/」API呼叫來更新檔案或目錄的特定安全性描述元資訊、例如主要擁有者、群組或控制旗標。

申請範例

```
curl -u admin:<PASSWORD> -X PATCH -d '{ \"control_flags\": \"32788\",
\"group\": \"everyone\", \"owner\": \"user1\"}' -k
'https://<IP_ADDRESS>/api/protocols/file-security/permissions/9479099d-
5b9f-11eb-9c4e-0050568e8682/%2Fparent?return_timeout=0'
```

回應範例

```
{
  "job": {
    "uuid": "6f89e612-5bbd-11eb-9c4e-0050568e8682",
    "_links": {
      "self": {
        "href": "/api/cluster/jobs/6f89e612-5bbd-11eb-9c4e-0050568e8682"
      }
    }
  }
}
```

刪除現有的SACL/DACL存取控制項目（ACE）

您可以使用「刪除/medias/file-security/permissions/」API呼叫、從目錄的檔案中刪除現有的ACE。此範例會將變更傳播至任何子物件。

申請範例

```
curl -u admin:<PASSWORD> -X DELETE -d '{ \"access\": \"access_allow\",
\"apply_to\": { \"files\": true, \"sub_folders\": true, \"this_folder\":
true }, \"ignore_paths\": [ \"/parent/child2\" ], \"propagation_mode\":
\"propagate\"}' -k 'https://<IP_ADDRESS>/api/protocols/file-
security/permissions/9479099d-5b9f-11eb-9c4e-
0050568e8682/%2Fparent/acl/himanshu?return_timeout=0'
```

回應範例

```
{
  "job": {
    "uuid": "e5683b61-5bbf-11eb-9c4e-0050568e8682",
    "_links": {
      "self": {
        "href": "/api/cluster/jobs/e5683b61-5bbf-11eb-9c4e-0050568e8682"
      }
    }
  }
}
```

不只是使用靜態API、更是使用非靜態CLI命令ONTAP ONTAP

利用此功能、您可以使用比支援許多工作的CLI更少的命令來自動化工作流程。ONTAP ONTAP例如、您可以使用單一POST API方法來修改檔案的安全性描述元、而非使用多個CLI命令。下表顯示完成一般檔案系統權限工作所需的CLI命令、以及對應的REST API呼叫：

靜態API ONTAP	CLI ONTAP
「Get /傳輸 協定/file-Security /有效權限/」	「vserver安全性檔案目錄show-aive-permissions」
「POST /傳輸協定/檔案安全性/權限/」	1. 「vserver安全性檔案目錄NTFS建立」 2. 「vserver安全檔案目錄NTFS DACL add」 3. 「vserver安全檔案目錄NTFS SACL add」 4. 「vserver安全性檔案目錄原則建立」 5. 「vserver安全性檔案目錄原則工作新增」 6. 「適用Vserver安全檔案目錄」
「修補程式/傳輸協定/檔案安全性/權限/」	「vserver安全性檔案目錄NTFS修改」
「刪除/傳輸協定/檔案安全性/權限/」	1. 「Vserver安全檔案目錄NTFS DACL移除」 2. 「Vserver安全檔案目錄NTFS SACL移除」

相關資訊

- "靜態API DACL/SACL權限範例Python指令碼ONTAP"
- "利用REST API簡化檔案安全權限的管理ONTAP"
- "搭配ONTAP 使用私有CLI passThrough搭配使用REST API（適用於ONTAP 版本9.9.1之前的版本）"

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